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N + - Network Fundamentals & Building Networks.

Introduction to VPNs.

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TASK 1.

❖ In Cisco Packet Tracer, create two separate networks (representing two branch offices) using two routers and configure basic IP addressing so that each network can ping its own devices.

- **Creating Two Separate Branch Office Networks:**

Network Setup: Two separate networks were created in Cisco Packet Tracer, representing two branch offices. Each network contains one router and a PC.

IP Address Configuration:

Branch Office	Device	IP Address	Subnet Mask
Branch 1	Router1 (LAN)	192.168.1.1	255.255.255.0
Branch 1	PC1	192.168.1.2	255.255.255.0
Branch 2	Router2 (LAN)	192.168.2.1	255.255.255.0
Branch 2	PC2	192.168.2.2	255.255.255.0

Connectivity Test: Ping tests were performed within each branch network.

PC1> ping 192.168.1.1
PC2> ping 192.168.2.1

Result: Both branch networks were configured successfully, and each PC could communicate with its own router.

TASK 2.

❖ Configure a basic site-to-site VPN between the two routers in Packet Tracer using GRE tunnel interfaces and assign tunnel IP addresses so that PCs from one branch can ping PCs in the other branch, Use the 'interface tunnel' command on both routers and set the tunnel source and destination to the routers' public-facing interfaces.

- **Configuring a Site to Site GRE VPN:**

Configure GRE Tunnel on Router1:

Router1(config)# interface tunnel 0
Router1(config-if)# ip address 10.10.10.1 255.255.255.252
Router1(config-if)# tunnel source 203.0.113.1
Router1(config-if)# tunnel destination 203.0.113.2

Configure GRE Tunnel on Router2:

Router2(config)# interface tunnel 0
Router2(config-if)# ip address 10.10.10.2 255.255.255.252
Router2(config-if)# tunnel source 203.0.113.2
Router2(config-if)# tunnel destination 203.0.113.1

Test Connectivity: A ping test was performed between PCs in both branch networks.

PC1> ping 192.168.2.2

Result: The GRE tunnel was configured between the two routers, allowing communication between the branch networks through the tunnel.

TASK 3.

❖ Simulate a remote access VPN scenario in Packet Tracer by configuring a VPN client PC to connect securely to one of the branch routers using the built-in VPN client feature, Use the VPN tab on the PC and set the correct server IP, username, and password.

- **Configuring a Remote Access VPN:**

VPN Configuration: The VPN client feature was opened on the remote PC in Cisco Packet Tracer.

Enter VPN Details:

Setting	Configuration
VPN Server IP	203.0.113.1
Username	vpnuser
Password	Vpn@12345

Connect VPN: The correct server IP, username, and password were entered, and the Connect option was selected.

Result: The remote PC was configured to connect to the branch router through a remote access VPN.

TASK 4.

❖ List and explain the main differences between site-to-site VPN and remote access VPN based on your Packet Tracer configurations, including at least two advantages and disadvantages of each approach.

- **Site to Site VPN vs. Remote Access VPN:**

Basis	Site-to-Site VPN	Remote Access VPN
1. Connection	Connects two or more office networks securely.	Connects an individual device to an office network securely.
2. Configuration	Configured between routers or VPN gateways.	Configured on individual devices using VPN client software.
3. Example	Connects Ahmedabad and Surat branch offices.	Allows an employee to access the office network from home.
4. Advantage 1	Provides continuous connectivity between branches.	Allows employees to work remotely from anywhere.
5. Advantage 2	Multiple users can communicate without setting up separate VPN connections.	Easy to configure for individual users.
6. Disadvantage 1	Requires VPN-capable routers or gateways at both locations.	Each user needs a separate VPN connection and configuration.
7. Disadvantage 2	Configuration and maintenance can be complex.	Performance depends on the user's internet connection.

Conclusion: Site to Site VPN is suitable for connecting branch offices, while Remote Access VPN is useful for employees who need secure access to office resources from remote locations.

TASK 5.

❖ Document and submit the step-by-step configuration commands you used to set up the site-to-site VPN in Packet Tracer, including interface, tunnel, and routing commands.

- **Step by Step Site to Site VPN Configuration:**

Configure Router1 Interfaces:

Router1(config)# interface g0/0
Router1(config-if)# ip address 192.168.1.1 255.255.255.0
Router1(config-if)# no shutdown
Router1(config)# interface g0/1
Router1(config-if)# ip address 203.0.113.1 255.255.255.252
Router1(config-if)# no shutdown

Configure Router2 Interfaces:

Router2(config)# interface g0/0
Router2(config-if)# ip address 192.168.2.1 255.255.255.0
Router2(config-if)# no shutdown
Router2(config)# interface g0/1
Router2(config-if)# ip address 203.0.113.2 255.255.255.252
Router2(config-if)# no shutdown

Configure GRE Tunnel on Router1:

Router1(config)# interface tunnel 0
Router1(config-if)# ip address 10.10.10.1 255.255.255.252
Router1(config-if)# tunnel source g0/1

Router1(config-if)# tunnel destination 203.0.113.2

Configure GRE Tunnel on Router2:

Router2(config)# interface tunnel 0
Router2(config-if)# ip address 10.10.10.2 255.255.255.252
Router2(config-if)# tunnel source g0/1
Router2(config-if)# tunnel destination 203.0.113.1

Configure Static Routes
On Router1:

Router1(config)# ip route 192.168.2.0 255.255.255.0 10.10.10.2

On Router2:

Router2(config)# ip route 192.168.1.0 255.255.255.0 10.10.10.1

Result: The GRE tunnel and static routes were configured between both routers, allowing devices in the Ahmedabad and Surat branch networks to communicate.